

Graph-Surrogate Quantum Architecture Search for Electronic-Structure and Correlated-Materials Hamiltonians

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Abstract

Variational quantum eigensolvers can approximate ground states relevant to molecular and correlated-material modelling, but their accuracy depends strongly on the selected circuit ansatz. We present a small graph-surrogate benchmark for ranking particle-number-preserving variational circuits before expensive optimization. The study includes a four-qubit, two-site spinful Fermi–Hubbard model at six interaction ratios U/t , together with six-qubit active-space LiH and BeH₂ Hamiltonians constructed in STO–3G and mapped by Jordan–Wigner. Twelve single- and double-excitation architectures were evaluated for each Hamiltonian, producing 96 VQE runs with two deterministic restarts.

Circuit graphs encode excitation type, position, and active spin orbitals, while Hamiltonian graphs summarize signed Pauli content and interaction-weighted qubit connectivity. A message-passing representation and validation-selected ridge surrogate were trained at $U/t = \{0.5, 2, 6\}$, tuned only at $U/t = 1$, and tested at the unseen values $U/t = \{3, 4\}$. The resulting rankings achieved Kendall correlations of 0.606 and 0.667. When only three finalists were fully evaluated, surrogate selection reduced the mean energy gap relative to random selection by 80% and 78%, respectively. The best LiH and BeH₂ circuits reached gaps of 7.1×10^{-5} and 4.9×10^{-5} hartree. Adding the current Hamiltonian descriptor did not improve within-family ranking, and zero-shot molecular transfer remained weaker ($\bar{\tau} = 0.288$). These results identify graph ranking as promising for interaction-parameter transfer while motivating learned Hamiltonian encoders for transferable materials simulation.

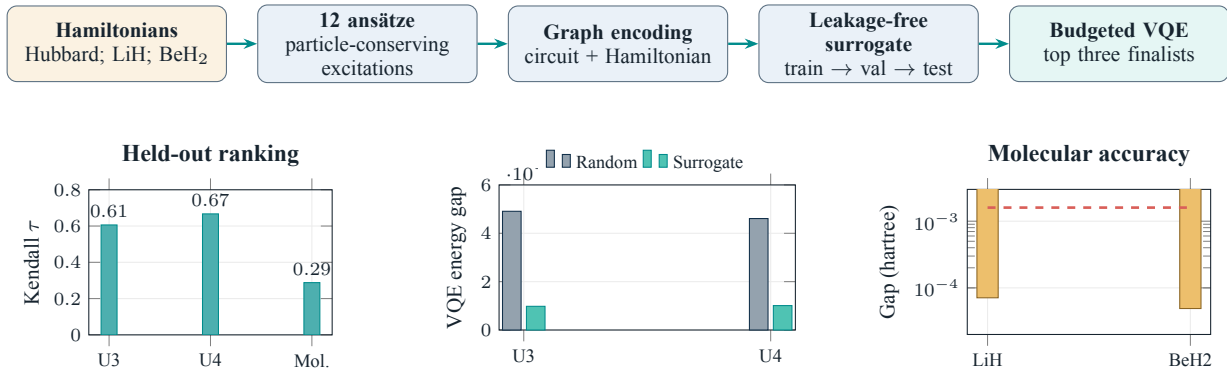


Figure 1. Hamiltonian-aware QAS workflow and held-out results. The red dashed line marks chemical accuracy ($\approx 1.6 \times 10^{-3}$ hartree).

Keywords: quantum architecture search; variational quantum eigensolver; electronic structure; Fermi–Hubbard model; graph surrogate

References. [1] Z. He et al., *Self-supervised representation learning for Bayesian quantum architecture search*, Phys. Rev. A, vol. 111, 032403, 2025. [2] S. Li et al., *Quantum architecture search with neural predictor based on ZX-calculus*, EPJ Quantum Technol., vol. 12, 106, 2025. [3] J. Chen et al., *Automated unitary coupled cluster circuit design via differentiable quantum architecture search*, arXiv:2605.28049, 2026. **Code:** github.com/SyedT1/Graph-Attention-QAS.